

Sam Ruttgaizer

Mechatronics Engineering - Memorial University of Newfoundland
St. John's, NL, Canada

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Skills

Programming: C, C++, Python, MATLAB, Julia, PlantUML

Embedded Systems: ESP-IDF, FreeRTOS, BLE, I2C, UART, RS-485, Real-time Signal Processing, EMG & IMU Sensors

Tools: SolidWorks, KiCad, Git, LaTeX, SPICE, MS Office, Oscilloscope, Logic Analyzer, Spectrum Analyzer

Professional Skills: Leadership, Team Communication, Problem Solving, Research

Scientific/MRI Software: SEPIA, SCT, ANTs, FSL, ITK-SNAP, ROMEO, STI Suite, qMRLab

Professional Experience

C-CORE

St. John's, NL

MUNSTAR-1 Satellite Mechatronics Engineering Student

May - August (2024) & September - December (2025)

- Built an end-to-end UHF radio communication system in C/Python, achieving a bit error rate $< 0.001\%$.
- Implemented serial communication protocols (I2C and UART) in C to allow seamless communication between the onboard computer, GPS, and Electrical Power System on the satellite.
- Developed radio state machine using PlantUML and Cogent allowing for seamless error handling and recovery during data transmission, improving overall system robustness.
- Maintained project repository in GitLab, reviewing merge requests and documenting software design decisions across a multi-disciplinary team.

Polytechnique Montréal

Montréal, QC

Research Intern, NeuroPoly Laboratory

April 2026 - August 2026

- Built and debugged an end-to-end quantitative MRI pipeline for ex vivo human spinal cord data using a mixed MATLAB/Python/Julia toolchain spanning phase unwrapping, background field removal, and B0 field mapping.
- Built an ANTs-based coregistration pipeline and performed statistical analysis validating a novel processing method against histological ground truth ($r = 0.64\text{--}0.66$, $p < 0.05$).

Student Teams

Memorial Medtech - Software Team Lead

2025 - Present

Biomedical engineering student team; won 1st place at the 2026 True North Biomedical Competition (beating 9 schools nationally) with a smart knee sleeve for ACL rehabilitation. Currently developing an emergency triage device.

- Currently leading development of a new portable wrist-worn triage device, applying PPG sensing to determine heart rate, rhythm, SpO2, and machine learning to estimate blood pressure.
- Wrote embedded firmware on ESP32 for EMG, IMU, and ROM sensor polling and real-time BLE packetization, using FreeRTOS to manage concurrent sampling, transmission, and logging tasks for live physiotherapy data streaming.
- Simulated and validated the EMG circuit design in PSPICE ahead of PCB design to optimize signal quality and minimize noise for accurate muscle activation readings.
- Directed development of a physiotherapist dashboard and managed technical direction for a 23-member interdisciplinary group, contributing to a 1st-place national finish among 10 competing universities.

Volunteering & Community

Easter Seals NL

St. John's, NL

Sledge Hockey Volunteer

2026

Instruct players through skating, puck handling, and shooting drills, creating an inclusive environment for all skill levels.

Burin Minor Soccer Association

Burin, NL

Soccer Coach

2023

Instructed players ages 3-15 on soccer fundamentals and organized weekend tournaments for 10+ teams.

Education

Memorial University of Newfoundland

St. John's, NL

Bachelor of Engineering (Co-op), Mechatronics Engineering Major

Class of 2028 GPA: 4.0, 92.3% Average

Dean's List: 2024, 2025, 2026

Research Science Institute – MIT, Boston, MA (2022)

Interests

Middle & Long-Distance Running, Hiking, Chess, Embedded Systems, Reading, Healthcare Innovations